1 A conjecture

this is a test Consider two energy eigenstates of a CFT $|\mathcal{O}_\alpha\rangle$ and $|\mathcal{O}_\beta\rangle$ corresponding to Virasoro primaries $\mathcal{O}_\alpha$ and $\mathcal{O}_\beta$ of dimensions h_α and h_β, respectively. Consider a subregion of angular size $A = (-\theta, \theta)$ on a spatial circle of circumference 2π, with $\theta = \pi x$. We denote by ρ_α and ρ_β the reduced states on A in their corresponding vectors.

The conjecture is that for the mixture $1/2(\rho_\alpha + \rho_\beta)$ we have the following asymptotic behavior of relative entropy

$$\begin{aligned} &S(x) = S(\rho_\alpha|(\rho_\alpha + \rho_\beta)/2) \\ &S(x) = A_0 x^{\delta_0}, \qquad S(1-x) = A_1(1-x)^{\delta_1}, \\ &\delta_1 \leq \delta_0 \ . \end{aligned} \tag{1.1}$$

In particular, we will be interested in the special case of massless 2d free boson CFT and energy eigenstates created by the action of vertex operator states $V_\alpha = e^{i\alpha\phi}$ and $V_\beta = e^{i\beta\phi}$ at the center of radial quantization.

2 Relative entropy for mixed excited states

Setup and Replica Trick

Consider a conformal field theory (CFT) in $1+1$ dimensions on a cylinder $\mathcal{C}_1$ of circumference R. We focus on a spatial subsystem A defined by the interval $(-l/2, l/2)$ at a fixed time slice. We simplify our notation by taking $A = (-\theta, \theta)$ on a cylinder of circumference 2π such that $\theta = \pi x$.

Our states of interest are the reduced density matrices corresponding to global excited coherent states. These states are generated by the action of vertex operators $V_\alpha =: e^{i\alpha\phi} :$ on the vacuum at past infinity in imaginary time on the Euclidean cylinder $\mathcal{C}_1$:

$$|\alpha\rangle = \lim_{u \to i\infty} \frac{V_\alpha(u)|\Omega\rangle}{\langle V_{-\alpha}(-u) V_\alpha(u)\rangle^{1/2}} \tag{2.1}$$

The reduced density matrix of the state $|\alpha\rangle$ on the subsystem A is denoted by ρ_α. A simple way to keep track of normalization of the density matrix to defined normalized operators

$$\tilde{V}_\alpha(u^+)\tilde{V}_{-\alpha}(u^-) = \frac{V_\alpha(u^+)V_{-\alpha}(u^-)}{\langle V_\alpha(u^+)V_{-\alpha}(u^-)\rangle_{\mathcal{C}_1}} \tag{2.2}$$

We aim to compute the relative entropy $S(\rho||\sigma_y)$ where $\rho = \rho_\alpha$ and the reference state is the mixture $\sigma_y = y\rho_\alpha + (1-y)\rho_\beta$ for $0 \leq y \leq 1$. The standard replica trick for relative entropy relates this quantity to the analytic continuation of trace functionals that break replica symmetry. Specifically, for the mixed state case, we utilize the following identity

derived from the generalized replica trick for normalized density matrices:

$$
\begin{aligned}
S\left(\rho_\alpha \middle\| y\rho_\alpha + (1-y)\rho_\beta\right) &= \partial_n \log\left[\frac{\mathrm{tr}(\rho_\alpha^n)}{\mathrm{tr}(\rho_\alpha(y\rho_\alpha + (1-y)\rho_\beta)^{n-1})}\right]_{n\to 1} \\
&= \partial_n \log\left[\frac{\langle \prod_{i=1}^n \tilde{V}_\alpha(u_i^+)\tilde{V}_{-\alpha}(u_i^-)\rangle_{\mathcal{C}_n}}{\langle \tilde{V}_\alpha(u_n^+)\tilde{V}_{-\alpha}(u_n^-)\mathcal{O}_{\alpha|\beta}^{(n-1)}\rangle_{\mathcal{C}_n}}\right]_{n=1}
\end{aligned}
\tag{2.3}
$$

where

$$
\mathcal{O}_{\alpha|\beta}^{(n-1)} = \sum_{p=0}^{n-1} y^{n-1-p}(1-y)^p \sum_{\substack{w\in\{\alpha,\beta\}^{n-1} \\ N_\beta(w)=p}} \tilde{V}_{w_1}(u_1^+)\tilde{V}_{-w_1}(u_1^-)\cdots\tilde{V}_{w_{n-1}}(u_{n-1}^+)\tilde{V}_{-w_{n-1}}(u_{n-1}^-)
\tag{2.4}
$$

and $w = (w_1, \ldots, w_{n-1})$ is an ordered word and $N_\beta(w)$ counts how many letters equal β, and the points are on the n-cover of cylinder $\mathcal{C}_n$ at $\pm i\infty$ on each sheet.

The conformal map

$$
z = \left(\frac{\sin((u-\theta)/2)}{\sin((u+\theta)/2)}\right)^{1/n}
\tag{2.5}
$$

sends the interval $A = (-\theta, \theta)$ to $A = (0, \infty)$ unifomizing $\mathcal{C}_n$ to the complex plane. Under this transformation Virasoro primaries transform as

$$
\begin{aligned}
\mathcal{O}(u,\bar{u}) &= \left(\frac{\partial z}{\partial u}\right)^{h_\alpha}\left(\frac{\partial \bar{z}}{\partial \bar{u}}\right)^{\bar{h}_\alpha}\mathcal{O}(z,\bar{z}) \\
V_\alpha(u_k,\bar{u}_k) &= \left(\frac{z_{k,n}\Lambda}{n}\right)^{h_\alpha}\left(\frac{\bar{z}_{k,n}\Lambda}{n}\right)^{\bar{h}_\alpha} V_\alpha(z_{k,n},\bar{z}_{k,n}),
\end{aligned}
\tag{2.6}
$$

where in our example

$$
z_{k,n}^{\pm} = e^{\frac{i\pi}{n}(2k\pm x)}, \qquad \Lambda = \frac{\sin(\theta)}{\cos(\theta)-\cos(u)}
\tag{2.7}
$$

Therefore, for normalized operators $\tilde{V}$ the contribution of Λ terms cancel and we find

$$
\begin{aligned}
\tilde{V}_\alpha(u_k,\bar{u}_k) &= \left(\frac{z_{k,n}}{nz_{1,1}}\right)^{h_\alpha}\left(\frac{\bar{z}_{k,n}}{nz_{1,1}}\right)^{\bar{h}_\alpha}\tilde{V}_\alpha(z_{k,n},\bar{z}_{k,n}) \\
&= \left(\frac{e^{i\pi(2k\pm x)(1-n)/n}}{n}\right)^{h_\alpha}\left(\frac{e^{-i\pi(2k\pm x)(1-n)/n}}{n}\right)^{\bar{h}_\alpha}\tilde{V}_\alpha(z_{k,n},\bar{z}_{k,n})
\end{aligned}
\tag{2.8}
$$

For scalar (spinless) operators in $2d$ we have $h_\alpha = \bar{h}_\alpha$ and we find

$$
\tilde{V}_\alpha(u_k,\bar{u}_k) = n^{-2h_\alpha}\tilde{V}_\alpha(z_{k,n},\bar{z}_{k,n})
\tag{2.9}
$$

Further assuming that $h_\alpha = h_{-\alpha}$ the relative entropy replica trick in terms of the correlator on the complex plane gives

$$
\begin{aligned}
&S(\rho_\alpha \| y\rho_\alpha + (1-y)\rho_\beta) = \partial_n \log \mathcal{F}_n\big|_{n\to 1} \\
&\mathcal{F}_n \equiv \left[\frac{\langle \prod_{k=1}^n \tilde{V}_\alpha(z_{k,n}^+)\tilde{V}_{-\alpha}(z_{k,n}^-)\rangle}{\langle \tilde{V}_\alpha(z_{n,n}^+)\tilde{V}_{-\alpha}(z_{n,n}^-)\tilde{\mathcal{O}}_{\alpha|\beta}^{(n-1)}\rangle} \right]_{n=1} \\
&\tilde{\mathcal{O}}_{\alpha|\beta}^{(n-1)} = \sum_{p=0}^{n-1} y^{n-1-p}(1-y)^p\, n^{4p(h_\alpha - h_\beta)} \sum_{\substack{w\in\{\alpha,\beta\}^{n-1} \\ N_\beta(w)=p}} \prod_{k=1}^{n-1} \frac{V_{w_k}(z_{k,n}^+)V_{-w_k}(z_{k,n}^-)}{\langle V_{\omega_k}(z_{1,1}^+)V_{-\omega_k}(z_{1,1}^-)\rangle}
\end{aligned}
\tag{2.10}
$$

Small x limit

To simplify our notation we define $C_\omega = C^L_{\omega(-\omega)}$ for $\omega = \alpha, \beta$. In the small x limit, we can expand in the OPE channel

$$
\frac{V_{w_k}(z_k^+)V_{-w_k}(z_k^-)}{\langle V_{w_k}(z^+)V_{-w_k}(z^-)\rangle} = \left(\frac{\sin(\pi x)}{\sin(\pi x/n)}\right)^{2h_{\omega_k}} \left(1 + C_\omega (2\sin(\pi x/n))^{\Delta_L}\mathcal{O}_L(e^{2\pi i k/n}) + \cdots\right)
$$

where $\mathcal{O}_L$ is the lightest Virasoro primary that we assume to have dimensions $(h_L, \bar{h}_L)$, weight $\Delta_L = h_L + \bar{h}_L$ less than two and spin $s_L = h_L - \bar{h}_L$ [1]. Therefore, we find

$$
\mathcal{F}_n = \frac{1 + (2\sin(\pi x/n))^{2\Delta_L} f(n) + \cdots}{\displaystyle\sum_{p=0}^{n-1} y^{n-1-p}(1-y)^p \left(\frac{n^2\sin(\pi x/n)}{\sin(\pi x)}\right)^{4p(h_\alpha - h_\beta)} \sum_{\substack{w\in\{\alpha,\beta\}^{n-1} \\ N_\beta(w)=p}} \left[1 + (2\sin(\pi x/n))^{2\Delta_L} g_w(n) + \cdots\right]}
\tag{2.11}
$$

where the leading correction $f(n)$ and $g_p(n)$ are given by the following sums over OPE coefficients and light operator correlators on the replica manifold:

$$
\begin{aligned}
f(n) &= C_\alpha^2 \sum_{k\neq k'=1}^{n} \langle \mathcal{O}_L(e^{2\pi i k/n})\mathcal{O}_L(e^{2\pi i k'/n})\rangle \\
&= C_\alpha^2 \frac{n}{2} \sum_{k=1}^{n-1} \frac{e^{-2\pi i s_L (k+k')/n}}{(2\sin(\pi k/n))^{2\Delta_L}} \ .
\end{aligned}
\tag{2.12}
$$

Rewrite the sum above by fixing the separation $m := k - k' \pmod n$. For each $m \in \{1, 2, \ldots, n-1\}$, set $k' = k - m \pmod n$. Then

$$
e^{-2\pi i s_L (k+k')/n} = e^{-2\pi i s_L (2k-m)/n} = e^{+2\pi i s_L m/n}\, e^{-4\pi i s_L k/n},
$$

[1] Otherwise the lowest order term would come from stress tensor.

and the denominator depends only on m. Hence the sum factorizes:

$$\sum_{m=1}^{n-1} \frac{e^{2\pi i s_L m/n}}{\left(2\sin\frac{\pi m}{n}\right)^{2\Delta_L}} \underbrace{\sum_{k=0}^{n-1} e^{-4\pi i s_L k/n}}_{=:G(n)} . \tag{2.13}$$

The inner sum $G(n)$ is a geometric series with ratio $q = e^{-4\pi i s_L/n}$:

$$G(n) = \sum_{k=0}^{n-1} q^k = \begin{cases} n, & q = 1, \\ 0, & q \neq 1. \end{cases}$$

Therefore $S(n) = 0$ for integer n unless $q = 1$, i.e.

$$e^{-4\pi i s_L/n} = 1 \quad \Longleftrightarrow \quad \frac{4s_L}{n} \in \mathbb{Z}.$$

Now, since $4s_L/n$ is not an integer, the numerator $e^{2\pi i s_L m/n}$ is at most a sign. For $s_k = 0$, we are left with

$$S_2(n) = \frac{n}{2} \sum_{k=1}^{n-1} (2\sin(\pi k/n))^{-2\Delta_L}, \qquad S_2(1) = 0, \tag{2.14}$$

whose analytic continuation of $S_2(n)$ was discussed in [**?**]

$$S_2'(1) = \frac{\sqrt{\pi}\Gamma(\Delta_L + 1)}{4\Gamma(\Delta_L + 3/2)} . \tag{2.15}$$

In a general CFT, there could be several inequivalent choices of the lightest primary operator $\mathcal{O}_L$ that contribute at the same order. In this case, the leading correction $f(n)$ would be given by the sum over all such operators, that following [**?**] we denote by

$$f(n) = \mathcal{N} S_2(n) \tag{2.16}$$

with $\mathcal{N} = 2$ for free boson and $\mathcal{N} = 1$ for Ising model. For similicity, we set $\mathcal{N} = 1$ in the remainder of this calculation.

Similarly, we have

$$g_w(n) = C_\alpha \sum_{k=1}^{n-1} \frac{C_{\omega_k}\mathcal{N}}{(2\sin(\pi k/n))^{2\Delta_L}} + \sum_{k\neq k'=1}^{n-1} \frac{C_{\omega_k} C_{\omega_{k'}}}{(2\sin(\pi|k-k'|/n))^{2\Delta_L}}, \tag{2.17}$$

We can simplify the denominator of $\mathcal{F}_n$ by noting that

$$\sum_{\substack{w\in\{\alpha,\beta\}^{n-1} \\ N_\beta(w)=p}} \left[1 + (2\sin(\pi x/n))^{2\Delta_L} g_w(n) + \cdots\right] = \binom{n-1}{p} + (2\sin(\pi x/n))^{2\Delta_L} \bar{g}_p(n) + \cdots$$

$$\bar{g}_p(n) = \sum_{\substack{w\in\{\alpha,\beta\}^{n-1} \\ N_\beta(w)=p}} g_w(n) = C_\alpha \sum_{k=1}^{n-1} \frac{A_k}{(2\sin(\pi k/n))^{2\Delta_L}} + \sum_{k\neq k'=1}^{n-1} \frac{B_{kk'}}{(2\sin(\pi|k-k'|/n))^{2\Delta_L}}, \tag{2.18}$$

where we have exchanged the order of the sum over words and the sum over replica indices and defined

$$
\begin{aligned}
A_k(p,n) &= \sum_{\substack{w\in\{\alpha,\beta\}^{n-1}\\ N_\beta(w)=p}} C_{\omega_k} = \binom{n-2}{p} C_\alpha + \binom{n-2}{p-1} C_\beta \equiv A(p,n) \\
B_{kk'}(p,n) &= \sum_{\substack{w\in\{\alpha,\beta\}^{n-1}\\ N_\beta(w)=p}} C_{\omega_k} C_{\omega_{k'}} = \binom{n-3}{p} C_\alpha^2 + 2\binom{n-3}{p-1} C_\alpha C_\beta + \binom{n-3}{p-2} C_\beta^2 \equiv B(p,n),
\end{aligned}
\tag{2.19}
$$

where we performed the sums over words for fixed $k \neq k'$ and found that the result is independent of k and k'. Therefore, we obtain

$$
\begin{aligned}
g_p(n) &= C_\alpha A(p,n) \sum_{k=1}^{n-1} \frac{1}{(2\sin(\pi k/n))^{2\Delta_L}} + B(p,n) \sum_{k\neq k'=1}^{n-1} \frac{1}{(2\sin(\pi|k-k'|/n))^{2\Delta_L}} \\
&= (C_\alpha A(p,n) - B(p,n)) \sum_{k=1}^{n} \frac{1}{(2\sin(\pi k/n))^{2\Delta_L}} + B(p,n) \sum_{k\neq k'=1}^{n} \frac{1}{(2\sin(\pi|k-k'|/n))^{2\Delta_L}} \\
&= \frac{2S_2(n)}{n}(C_\alpha A(p,n) - B(p,n)) + B(p,n) S_2(n)
\end{aligned}
$$

and as a result we can write

$$
\left[\, 1 + (2\sin(\pi x/n))^{2\Delta_L} g_w(n) + \cdots \right] = \binom{n-1}{p} + (2\sin(\pi x/n))^{2\Delta_L} g_p(n) + \cdots \tag{2.20}
$$

Defining

$$
D_n = \left(\frac{n^2 \sin(\pi x/n)}{\sin(\pi x)} \right)^{2(h_\alpha - h_\beta)}, \qquad A_1 = 1, \tag{2.21}
$$

we note the following identities

$$Z(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p}((1-y)D_n)^p \binom{n-1}{p} = (y+(1-y)D_n)^{n-1} \tag{2.22}$$

$$A_\alpha(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p}((1-y)D_n)^p \binom{n-2}{p} = y(y+(1-y)D_n)^{n-2} \tag{2.23}$$

$$A_\beta(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p}((1-y)D_n)^p \binom{n-2}{p-1} = (1-y)D_n(y+(1-y)D_n)^{n-2} \tag{2.24}$$

$$B_{\alpha\alpha}(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p}((1-y)D_n)^p \binom{n-3}{p} = y^2(y+(1-y)D_n)^{n-3} \tag{2.25}$$

$$B_{\alpha\beta}(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p}((1-y)D_n)^p \binom{n-3}{p-1} = y(1-y)D_n(y+(1-y)D_n)^{n-3} \tag{2.26}$$

$$B_{\beta\beta}(n) \equiv \sum_{p=0}^{n-1} y^{n-1-p}((1-y)D_n)^p \binom{n-3}{p-2} = (1-y)^2D_n^2(y+(1-y)D_n)^{n-3}, \tag{2.27}$$

For simplicity, we define

$$\begin{aligned} X(n) &= (y+(1-y)D_n), \qquad X(1)=1 \\ Y_{\alpha\beta}(n) &= (yC_\alpha + (1-y)D_nC_\beta) \end{aligned} \tag{2.28}$$

so that

$$\begin{aligned} B(p,n) &= X(n)^{n-3}\left(C_\alpha^2 B_{\alpha\alpha}(n) + 2C_\alpha C_\beta B_{\alpha\beta}(n) + C_\beta^2 B_{\beta\beta}(n)\right) \\ &= Y_{\alpha\beta}(n)^2 X(n)^{n-3} \\ A(p,n) &= C_\alpha A_\alpha(n) + C_\beta A_\beta(n) = Y_{\alpha\beta}(n)X(n)^{n-2} \end{aligned} \tag{2.29}$$

Then, in the denominator we have

$$X(n)^{n-1}\left(1+\left(\frac{2\pi x}{n}\right)^{2\Delta_L}\Delta(n)+\cdots\right) \tag{2.30}$$

where

$$\begin{aligned} \Delta(n) &\equiv \frac{2S_2(n)}{n}\frac{(C_\alpha A(p,n) - B(p,n))}{X(n)^{n-1}} + \frac{B(p,n)}{X(n)^{n-1}}S_2(n) \\ &= \frac{2S_2(n)}{n}\left(\frac{C_\alpha Y_{\alpha\beta}(n)}{X(n)} - \frac{Y_{\alpha\beta}(n)^2}{X(n)^2}\right) + S_2(n)\left(\frac{Y_{\alpha\beta}(n)^2}{X(n)^2}\right) \qquad (2.31) \\ &= S_2(n)\left(\frac{Y_{\alpha\beta}(n)}{X(n)}\left(\frac{2C_\alpha}{n}\right) + \frac{Y_{\alpha\beta}^2(n)}{X(n)^2}\left(1-\frac{2}{n}\right)\right) \qquad (2.32) \end{aligned}$$

Putting this back into (2.11) we find

$$\log \mathcal{F}_n = (1-n)\log X(n) + \log\left[\frac{1+\left(\frac{2\pi x}{n}\right)^{2\Delta_L} S_2(n)\left(C_\alpha^2+\cdots\right)}{1+\left(\frac{2\pi x}{n}\right)^{2\Delta_L}\Delta(n)+\cdots}\right] \tag{2.33}$$

We first note that taking the n-derivative of the first factor in $\mathcal{F}_n$ and sending $n \to 1$ vanishes

$$\partial_n\left[(1-n)\log(y+(1-y)D_n)\right]\Big|_{n=1} = 0 \tag{2.34}$$

and we are left with

$$S(\rho_\alpha \| y\rho_\alpha + (1-y)\rho_\beta) = \partial_n\left[\left(\frac{2\pi x}{n}\right)^{2\Delta_L}\left(S_2(n)C_\alpha^2 - \Delta(n)\right)+\cdots\right]_{n=1} \tag{2.35}$$

Since $S_2(1)=0$ and $X(1)=1$, we have

$$\begin{aligned} S(\rho_\alpha \| y\rho_\alpha + (1-y)\rho_\beta) &= (2\pi x)^{2\Delta_L} S_2'(1)(C_\alpha^2 - 2C_\alpha Y_{\alpha\beta}(1) + Y_{\alpha\beta}(1)^2)+\cdots \\ &= (2\pi x)^{2\Delta_L}\frac{\sqrt{\pi}\Gamma(\Delta_L+1)}{4\Gamma(\Delta_L+3/2)}(1-y)^2(C_\alpha - C_\beta)^2+\cdots \end{aligned} \tag{2.36}$$

where in the second line we have used $S_2(1)=0$. In the special case, $\alpha=\beta$ or $y\to 1$ the term vanishes as expected. In the case of vertex operators in free massless bosons, we have

$$V_\omega(z,\bar z)V_{-\omega}(0,0) = |z|^{-2\Delta_\omega}\left(1 + i\omega\left(zJ(0)+\bar z\bar J(0)\right) + \omega^2(z\bar z J(0)\bar J(0)+\cdots)\right) \tag{2.37}$$

Small $1-x$ limit

The final answer will be (2.83). To start simple, we take $\alpha=0$ so that ρ_α is the vacuum state reduced to the subsystem.

The most non-trivial component of this calculation is the term $\mathrm{tr}\big(\rho_\alpha\,(y\rho_\alpha+(1-y)\rho_\beta)^{n-1}\big)$. Since the operators do not commute in general, one should expand the *power* as a sum over all words of length $n-1$ in the alphabet $\{\rho_\alpha,\rho_\beta\}$ rather than using a naive binomial theorem. In the case $\alpha=0$ we write ρ_0 for the vacuum reduced density matrix and obtain the precise finite-n identity

$$\mathrm{tr}\big(\rho_0\,(y\rho_0+(1-y)\rho_\beta)^{n-1}\big) = \sum_{p=0}^{n-1} y^{\,n-1-p}(1-y)^p \sum_{\substack{w\in\{0,\beta\}^{n-1}\\ N_\beta(w)=p}} \mathrm{tr}\big(\rho_0\,\rho_{w_1}\rho_{w_2}\cdots\rho_{w_{n-1}}\big)\,, \tag{2.38}$$

where $w=(w_1,\ldots,w_{n-1})$ is an ordered word and $N_\beta(w)$ counts how many letters equal β.

Equivalently, one may separate the "naive binomial" term (in which all ρ_β are taken to sit to the right) plus an explicit remainder built from commutators. A convenient way to package this remainder is via the adjoint action $\mathrm{ad}_{\rho_0}(X)\equiv[\rho_0,X]$:

$$\mathrm{tr}\big(\rho_0\,(y\rho_0+(1-y)\rho_\beta)^{n-1}\big) = \sum_{p=0}^{n-1}\binom{n-1}{p} y^{\,n-1-p}(1-y)^p\,\mathrm{tr}\Big(\rho_0^{\,n-p}\rho_\beta^{\,p}\Big) + \mathcal{C}_n(y), \tag{2.39}$$

with the commutator correction

$$\mathcal{C}_n(y) \equiv \sum_{p=0}^{n-1} y^{n-1-p}(1-y)^p \sum_{\substack{w\in\{0,\beta\}^{n-1}\\ N_\beta(w)=p}} \mathrm{tr}(\rho_0\,\Delta_w)\,, \qquad \Delta_w \equiv \rho_{w_1}\cdots\rho_{w_{n-1}} - \rho_0^{n-1-p}\rho_\beta^p. \quad (2.40)$$

Each Δ_w can be reduced to a finite sum of nested commutators by repeatedly commuting ρ_0 past ρ_β (so $\mathcal{C}_n(y)$ vanishes whenever $[\rho_0,\rho_\beta]=0$; in particular, for the free boson vertex-operator states considered below the ordering dependence drops out).

Using the operator-state correspondence, the trace $\mathrm{tr}(\rho_\alpha^{n-p}\rho_\beta^p)$ corresponds to a path integral on an n-sheeted Riemann surface with specific operator insertions. In the limit $n\to 1$, we can identify this with the correlation function of p operators of type β and $q=n-p$ operators of type α:

$$C_{q,p} \equiv \langle \mathcal{O}_\alpha^{(q)}\mathcal{O}_\beta^{(p)}\rangle \quad (2.41)$$

As was pointed out, in the case of free bosons, the order of operators will not matter, and the commutator terms vanish.

Operator Insertions on the Strip

We perform the calculation on a strip of width 2π, obtained via the standard conformal mapping from the n-sheeted cylinder. The replica sheets are indexed by $k=0,\ldots,n-1$.

The operator insertion points $t_k^\pm$ on the strip are:

$$t_k^\pm = \frac{2\pi k}{n} \pm \frac{\pi x}{n} \quad (2.42)$$

where $x=l/R$ is the dimensionless interval size. For the correlator $\langle \mathcal{O}_\alpha^{(q)}\mathcal{O}_\beta^{(p)}\rangle$ with $q=n-p$, the operators are explicitly distributed as follows: On sheets $k=0,\ldots,q-1$ (α-sector) we insert V_α at t_k^+ and $V_{-\alpha}$ at t_k^-, and on sheets $k=q,\ldots,n-1$ (β-sector) we insert V_β at t_k^+ and $V_{-\beta}$ at t_k^-.

Special case

For simplicity, focus on the special case $y=1/2$ so that we need to compute

$$\begin{aligned} 2^{-(n-1)}\frac{\mathrm{tr}(\rho_0(\rho_0+\rho_\beta)^{n-1})}{\mathrm{tr}(\rho_0^n)} &= 2^{-(n-1)}\Bigg(1+(n-1)\frac{\mathrm{tr}(\rho_0^{n-1}\rho_\beta)}{\mathrm{tr}(\rho_0^n)} + (n-1)\sum_{k=1}^{n-2}\frac{\mathrm{tr}(\rho_0^k\rho_\beta\rho_0^{n-1-k}\rho_\beta)}{\mathrm{tr}(\rho_0^n)} \\ &\quad +(n-1)\sum_{k_1=1}^{n-3}\sum_{k_2=0}^{n-3-k_1}\frac{\mathrm{tr}(\rho_0^{k_1}\rho_\beta\rho_0^{k_2}\rho_\beta\rho_0^{n-3-k_1-k_2}\rho_\beta)}{\mathrm{tr}(\rho_0^n)} + \cdots\Bigg) \end{aligned} \quad (2.43)$$

In this case, the first term $p=1$ gives

$$\begin{aligned} &\frac{\mathrm{tr}(\rho_0^{n-1}\rho_\beta)}{\mathrm{tr}(\rho_0^n)} = f(x)^{-\beta^2} \\ &f(x) \equiv (2\sin(\pi x/n)), \end{aligned} \quad (2.44)$$

whereas the $p = 2$ becomes

$$\sum_{k=1}^{n-2} \frac{\mathrm{tr}(\rho_0^k \rho_\beta \rho_0^{n-1-k} \rho_\beta)}{\mathrm{tr}(\rho_0^n)} = f_1(x)^{-2\beta^2} g_2(x;\beta)$$

$$g_2(x;\beta) = \sum_{k=1}^{n-2} \left[\frac{\sin^2\left(\frac{\pi k}{n}\right)}{\sin\left(\frac{\pi(k+x)}{n}\right) \sin\left(\frac{\pi(k-x)}{n}\right)} \right]^{\beta^2} . \tag{2.45}$$

For the $p = 3$ term, we find

$$\sum_{k_1=1}^{n-3} \sum_{k_2=0}^{n-3-k_1} \frac{\mathrm{tr}(\rho_0^{k_1} \rho_\beta \rho_0^{k_2} \rho_\beta \rho_0^{n-3-k_1-k_2} \rho_\beta)}{\mathrm{tr}(\rho_0^n)} = \cdots \tag{2.46}$$

Calculation of the correlator

The correlation function for primary vertex operators $V_Q =: e^{iQ\phi} :$ is given by the product of distances raised to the power of the product of their charges. The chordal distance on the strip is $d(t_i, t_j) = 2\sin(\frac{t_i - t_j}{2})$.

$$\langle \prod_i V_{Q_i}(t_i) \rangle \sim \prod_{i<j} \left[2\sin\left(\frac{t_i - t_j}{2}\right) \right]^{Q_i Q_j} \tag{2.47}$$

- **Like charges** ($Q_i Q_j > 0$): The operators repel (regular zero), leading to a positive exponent $+Q_i Q_j$.
- **Opposite charges** ($Q_i Q_j < 0$): The operators attract (singular pole), leading to a negative exponent $-|Q_i Q_j|$.

We have the following contributions:

1. Nearest Neighbor Interactions (Same sheet, same sector) These are the contractions between t_k^+ and t_k^- on the same sheet.

- α-sector (q pairs): $(\alpha, -\alpha)$ separated by $\Delta t = 2\pi x/n$.

$$\Longrightarrow \left(2\sin\frac{\pi x}{n}\right)^{-q\alpha^2}$$

- β-sector (p pairs): $(\beta, -\beta)$ separated by $\Delta t = 2\pi x/n$.

$$\Longrightarrow \left(2\sin\frac{\pi x}{n}\right)^{-p\beta^2}$$

Putting the two together, we have

$$\mathcal{N}(q, p; x) = \left(2\sin\frac{\pi x}{n}\right)^{-(q\alpha^2 + p\beta^2)} \tag{2.48}$$

2. Intra-sector Interactions (Different sheets, same sector) For the α-sector, we sum over sheet separations $k = 1, \ldots, q-1$. The number of pairs at distance k is $(q-k)$.

- Like charges (α, α) and $(-\alpha, -\alpha)$:

$$\text{Dist} = \frac{2\pi k}{n} \implies \text{Exp} = +\alpha^2 \times 2$$

- Opposite charges $(\alpha, -\alpha)$: Two diagonals $\frac{2\pi k}{n} \pm \frac{2\pi x}{n}$.

$$\text{Dist} = \frac{2\pi(k \pm x)}{n} \implies \text{Exp} = -\alpha^2 \text{ (each)}$$

Collecting these terms for the α sector gives $[g(q;x)]^{\alpha^2}$:

$$g(q;x) = \prod_{k=1}^{q-1} \left(\frac{\sin^2(\frac{\pi k}{n})}{\sin(\frac{\pi(k+x)}{n}) \sin(\frac{\pi(k-x)}{n})} \right)^{q-k} . \tag{2.49}$$

The β sector is identical with $q \to p$ and $\alpha \to \beta$, and we have the overall contribution

$$g(q;x)^{\alpha^2} g(p;x)^{\beta^2} . \tag{2.50}$$

3. Cross-sector Interactions Interactions between the q α-sheets and p β-sheets yield terms involving the product $\alpha\beta$. These collect into $[h(p,q;x)]^{\alpha\beta}$:

$$h(q,p;x) = \prod_{l=1}^{p} \prod_{k=1}^{q} \frac{\sin^2(\frac{\pi(k+l-1)}{n})}{\sin(\frac{\pi(k+l-1+x)}{n}) \sin(\frac{\pi(k+l-1-x)}{n})} \tag{2.51}$$

Putting all the pieces together, we find that the full correlator in (2.41) is

$$\langle \mathcal{O}_\alpha^{(q)} \mathcal{O}_\beta^{(p)} \rangle = \mathcal{N}(n-p,p;x) \, [g(n-p;x)]^{\alpha^2} \, [g(p;x)]^{\beta^2} \, [h(n-p,p;x)]^{\alpha\beta} . \tag{2.52}$$

We need to continue in n analytically:

$$F(n) \equiv \text{tr}(\rho(\rho+\sigma)^{n-1}) = \sum_{p=0}^{n-1} \binom{n-1}{p} \langle \mathcal{O}_\alpha^{(q)} \mathcal{O}_\beta^{(p)} \rangle \tag{2.53}$$

2.1 Analytic continuation in n

We consider the relative entropy $S(\rho_\alpha || \sigma)$ where $\sigma = y\rho_\alpha + (1-y)\rho_\beta$ for excited coherent states in a 1+1D free boson CFT. The replica trick requires evaluating:

$$\text{tr}(\rho_\alpha \sigma^{n-1}) = \sum_{p=0}^{n-1} \binom{n-1}{p} y^{n-1-p} (1-y)^p C_{n-p,p} \tag{2.54}$$

where $C_{n-p,p} = \langle \mathcal{O}_\alpha^{(n-p)} \mathcal{O}_\beta^{(p)} \rangle$ is the mixed correlator on the n-sheeted replica surface.

Step 1 Log-Sum expansion of the correlator: ,k,1,The correlator is a product of geometric functions g and h. We focus on the intra-sector term $\log g(p;x)$. For integer p, it is defined as:

$$\log g(p;x) = \sum_{k=1}^{p-1}(p-k)\left[2\log d\left(\frac{k}{n}\right) - \log d\left(\frac{k+x}{n}\right) - \log d\left(\frac{k-x}{n}\right)\right] \tag{2.55}$$

where $d(y) = 2\sin(\pi y)$ is the distance between operator insertions. We utilize the Fourier expansion for $\log\sin(\pi y)$:

$$\log\sin(\pi y) = -\log 2 - \sum_{m=1}^{\infty}\frac{\cos(2\pi m y)}{m} \tag{2.56}$$

Substituting this into the potential $\mathcal{V}(k,x)$ inside the sum, the constants cancel, yielding:

$$\mathcal{V}(k,x) = -2\sum_{m=1}^{\infty}\frac{1}{m}\left[1-\cos\left(\frac{2\pi m x}{n}\right)\right]\cos\left(\frac{2\pi m k}{n}\right) \tag{2.57}$$

The expression above has the advantage that it has separated the contribution of k from those of x, and we can now perform the sum over k.

Step 2 Sum over k: Following Section 6 of Calabrese, Cardy, and Tonni (2011), we perform the sum over k for each mode m:

$$\log g(p;x) = -2\sum_{m=1}^{\infty}\frac{1-\cos(2\pi m x/n)}{m}\sum_{k=1}^{p-1}(p-k)\cos\left(\frac{2\pi m k}{n}\right) \tag{2.58}$$

The inner sum $S_m(p) = \sum_{k=1}^{p-1}(p-k)\cos(k\theta_m)$ is evaluated using geometric series. For integer p and $\theta_m = 2\pi m/n$:

$$S_m(p) = \mathrm{Re}\left[\frac{pe^{i\theta_m}}{1-e^{i\theta_m}} - \frac{e^{i\theta_m}(1-e^{ip\theta_m})}{(1-e^{i\theta_m})^2}\right] = \frac{-p}{2} + \frac{(1-\cos(p\theta_m))}{4\sin^2(\theta_m/2)}, \tag{2.59}$$

Therefore, we find

$$\log g(p;x) = \sum_{m=1}^{\infty}\frac{1-\cos(\frac{2\pi m x}{n})}{m}\left(p - \frac{\sin^2(\pi p m/n)}{\sin^2(\pi m/n)}\right) \tag{2.60}$$

Define the mode kernels

$$L(n,x) := \sum_{m=1}^{\infty}\frac{1-\cos(2\pi m x/n)}{m}, \qquad K_m(n,x) := \frac{1-\cos(2\pi m x/n)}{2m\,\sin^2(\pi m/n)}. \tag{2.61}$$

to write

$$\log g(p;x) = p\,L(n,x) \;-\; \sum_{m=1}^{\infty}K_m(n,x)\,\big(1-\cos(p\theta_m)\big). \tag{2.62}$$

All p-independent pieces can be absorbed into $\Omega_n^{(\mathrm{rest})}(x)$ later.

A technical note to keep in mind is that to make every rearrangement fully justified, one may define all mode sums with a common Abel factor $e^{-\varepsilon m}$ in the summand and take $\varepsilon \to 0^+$ at the end. This guarantees absolute and uniform convergence in intermediate steps.

Next, we move to the function $h(q,p;x)$ in (2.51) which includes a double product over (k,l) with $1 \le k \le q$, $1 \le l \le p$. Taking logs and using the same log-sum representation as in we did for the g function yields

$$\log h(q,p;x) = \sum_{r=1}^{n-1} w_r(p,q)\, V(r,x), \tag{2.63}$$

where $w_r(p,q)$ counts the number of pairs (k,l) such that $k+l-1=r$.

It is convenient to evaluate the weighted cosine sum via generating functions. Let θ be arbitrary and note

$$\sum_{r=1}^{n-1} w_r(p,q) e^{\mathrm{i}r\theta} = \sum_{k=1}^{q}\sum_{l=1}^{p} e^{\mathrm{i}(k+l-1)\theta} = e^{-\mathrm{i}\theta}\Big(\sum_{k=1}^{q} e^{\mathrm{i}k\theta}\Big)\Big(\sum_{l=1}^{p} e^{\mathrm{i}l\theta}\Big). \tag{2.64}$$

Now specialize to the case $q=n-p$ and $\theta=\theta_m=2\pi m/n$. Using $\sum_{k=1}^{n-p} e^{\mathrm{i}k\theta_m} = -e^{-\mathrm{i}p\theta_m/2}\,\frac{\sin(p\theta_m/2)}{\sin(\theta_m/2)}$ and $\sum_{l=1}^{p} e^{\mathrm{i}l\theta_m} = e^{\mathrm{i}(p+1)\theta_m/2}\,\frac{\sin(p\theta_m/2)}{\sin(\theta_m/2)}$, the right-hand side of (2.64) becomes real and equals

$$\sum_{r=1}^{n-1} w_r(p,n-p)\cos(r\theta_m) = -\left(\frac{\sin(p\theta_m/2)}{\sin(\theta_m/2)}\right)^2 = -\frac{1-\cos(p\theta_m)}{2\sin^2(\theta_m/2)}. \tag{2.65}$$

Substituting (2.65) into (2.63) gives the exact mode form

$$\log h(n-p,p;x) = 2\sum_{m=1}^{\infty} K_m(n,x)\,\big(1-\cos(p\theta_m)\big), \tag{2.66}$$

with the *same* $K_m(n,x)$ as in (2.61).

We are now ready to put together all the pieces back into $C_{n-p,p}$. Using (2.62) and (2.66), we find

$$\begin{aligned}\alpha^2\log g(q;x) + \beta^2\log g(p;x) + \alpha\beta\log h(q,p;x) &= (\alpha^2 q + \beta^2 p)\,L(n,x)\\ &\quad - (\alpha-\beta)^2\sum_{m=1}^{\infty} K_m(n,x)\,\big(1-\cos(p\theta_m)\big)\end{aligned} \tag{2.67}$$

where we have used a similar log-sum expansion for the function $\mathcal{N}$. Absorbing all p-independent constants (including the $-(\alpha-\beta)^2\sum_m K_m$ term) into $\Omega_n(x)$, we obtain

$$C_{n-p,p}(x) = \Omega_n(x)\,\Lambda_\alpha^{n-p}\,\Lambda_\beta^{p}\,\exp\left[\sum_{m=1}^{\infty} a_m(n,x)\cos(p\theta_m)\right], \tag{2.68}$$

where

$$a_m(n,x) = (\alpha-\beta)^2\,K_m(n,x) = (\alpha-\beta)^2\,\frac{1-\cos(2\pi m x/n)}{2m\,\sin^2(\pi m/n)}\ . \tag{2.69}$$

2.2 Fourier/Bessel expansion and definition of the R-modes

Let $\theta := \frac{2\pi p}{n}$ so that $\cos(p\theta_m) = \cos(m\theta)$. Using $e^{a\cos(m\theta)} = \sum_{r\in\mathbb{Z}} I_r(a)\, e^{\mathrm{i}rm\theta}$ (modified Bessel functions), we expand

$$\exp\left[\sum_{m\geq 1} a_m \cos(m\theta)\right] = \sum_{R\in\mathbb{Z}} W_R(n,x;\alpha,\beta)\, e^{\mathrm{i}R\theta}, \tag{2.70}$$

where the coefficients are the (infinite) Bessel convolution

$$W_R(n,x;\alpha,\beta) = \sum_{\{r_m\}:\sum_{m\geq 1} m r_m = R} \prod_{m\geq 1} I_{r_m}\big(a_m(n,x)\big), \qquad W_{-R} = W_R. \tag{2.71}$$

This allows us to rewrite

$$C_{n-p,p}(x) = \Omega_n\, \Lambda_\alpha^{n-p}\Lambda_\beta^{p} \sum_{R\in\mathbb{Z}} W_R(n)\, e^{\mathrm{i}\frac{2\pi R}{n}p}. \tag{2.72}$$

Note that with an Abel regulator in the defining mode sums, the function of θ in (2.70) is real-analytic in a complex strip, which implies exponential decay of W_R and in particular $\sum_R |W_R| < \infty$.

The expression in (2.72) has the advantage that if we can swap the sum over R with the sum over p, we can perform the p-sum explicitly. That is our next step. **I believe swapping the two sums can be justified using Abel regulators but I am not entirely sure!**

Plugging these back in 2.53 and swapping the order of summation, we obtain

$$\begin{aligned} F(n) &= \Omega_n\, \Lambda_\alpha \sum_{R\in\mathbb{Z}} W_R(n) \sum_{p=0}^{n-1} \binom{n-1}{p} (y\Lambda_\alpha)^{n-1-p} \left((1-y)\Lambda_\beta e^{\mathrm{i}2\pi R/n}\right)^p \\ &= \Omega_n\, \Lambda_\alpha \sum_{R\in\mathbb{Z}} W_R(n) \left(y\Lambda_\alpha + (1-y)\Lambda_\beta e^{\mathrm{i}2\pi R/n}\right)^{n-1}. \end{aligned} \tag{2.73}$$

Next, we would like to argue that only the $R=0$ term in the expression above contrinutes to the $n\to 1$ limit. To establish this, define $\varepsilon = n-1$ and

$$A_R(n) := y\Lambda_\alpha + (1-y)\Lambda_\beta e^{\mathrm{i}2\pi R/n}, \qquad A_0 := y\Lambda_\alpha + (1-y)\Lambda_\beta > 0. \tag{2.74}$$

Then (2.73) reads

$$F(n) = \Omega_n \Lambda_\alpha \sum_{R\in\mathbb{Z}} W_R(n)\, A_R(n)^{\varepsilon}. \tag{2.75}$$

With an Abel regulator W_R decays exponentially and has finite moments. Expand the phase for $n = 1+\varepsilon$:

$$e^{\mathrm{i}2\pi R/n} = e^{\mathrm{i}2\pi R/(1+\varepsilon)} = \exp\big(-\mathrm{i}2\pi R\varepsilon + O(R\varepsilon^2)\big), \tag{2.76}$$

so $A_R(n) = A_0 + \delta_R$ with $\delta_R = O(R\varepsilon)$ for fixed R and $\delta_{-R} = \overline{\delta_R}$. Using

$$A_R^\varepsilon = \exp\big(\varepsilon \log(A_0 + \delta_R)\big) = A_0^\varepsilon \left[1 + \varepsilon \frac{\delta_R}{A_0} + O(\varepsilon|\delta_R|^2)\right], \tag{2.77}$$

and the evenness $W_{-R} = W_R$, the potentially dangerous linear term $\sum_R W_R\, \delta_R$ cancels at $O(\varepsilon)$ (its leading part is odd/purely imaginary), leaving only $O(\varepsilon^2)$ corrections after summing over R. Consequently, after absorbing $\sum_R W_R$ into Ω_n (a p-independent normalization),

$$F(1+\varepsilon) = \Omega_{1+\varepsilon} \Lambda_\alpha\, A_0^\varepsilon \big(1 + O(\varepsilon^2)\big). \tag{2.78}$$

Thus the $R \neq 0$ modes do not affect $\partial_n \log F(n)$ at $n = 1$.

The replica prescription for the relative entropy is

$$S(\rho_\alpha \| \sigma_y) = \partial_n \Big(\log \operatorname{tr}(\rho_\alpha^n) - \log \operatorname{tr}(\rho_\alpha \sigma_y^{n-1})\Big)\Big|_{n=1}. \tag{2.79}$$

In the same framework one has $\operatorname{tr}(\rho_\alpha^n) = \widetilde{\Omega}_n\, \Lambda_\alpha^n$ so that $\partial_n \log \operatorname{tr}(\rho_\alpha^n)|_{n=1} = \log \Lambda_\alpha$ (all $\widetilde{\Omega}_n$ terms cancel against Ω_n in the difference). Using (2.78),

$$\partial_n \log \operatorname{tr}(\rho_\alpha \sigma_y^{n-1})\big|_{n=1} = \log A_0 = \log\big(y\Lambda_\alpha + (1-y)\Lambda_\beta\big). \tag{2.80}$$

Therefore

$$S(\rho_\alpha \| \sigma_y) = \log \Lambda_\alpha - \log\big(y\Lambda_\alpha + (1-y)\Lambda_\beta\big) = -\log\left[y + (1-y)\frac{\Lambda_\beta}{\Lambda_\alpha}\right]. \tag{2.81}$$

For coherent-state excitations in the free boson CFT one may identify

$$\frac{\Lambda_\beta}{\Lambda_\alpha} = e^{-S(\alpha\|\beta)}, \qquad S(\alpha\|\beta) = (\alpha-\beta)^2 K(x), \qquad K(x) = 1 - \pi x \cot(\pi x). \tag{2.82}$$

Substituting (2.82) into (2.81) gives the final answer

$$\boxed{S\Big(\rho_\alpha \,\Big\|\, y\rho_\alpha + (1-y)\rho_\beta\Big) = -\log\big(y + (1-y)\exp\big[-(\alpha-\beta)^2(1-\pi x\cot(\pi x))\big]\big)\,.} \tag{2.83}$$

3 Consistency tests

Let $s := (\alpha-\beta)^2 K(x) \geq 0$ and $a := e^{-s} \in (0,1]$ so that

$$S(y) \equiv S\Big(\rho_\alpha \,\Big\|\, y\rho_\alpha + (1-y)\rho_\beta\Big) = -\log\left(y + (1-y)a\right) = -\log(a + (1-y)a) \tag{3.1}$$

We are gonna test the following limits:

- **Identical states:** $\alpha = \beta \Rightarrow s = 0 \Rightarrow a = 1 \Rightarrow S(y) = 0$ as expected.
- **No mixing:** $S(1) \to 0$ as expected and $S(0) = S(\alpha\|\beta)$ as expected. Therefore, we can write

$$e^{-S(y)} = y e^{-S(1)} + (1-y) e^{-S(0)}\,. \tag{3.2}$$

- **Much heavier state:** $s \to \infty \Rightarrow a \to 0 \Rightarrow S_y \to -\log y$.

Next, we test the following properties:

- **Positivity:** Since $y \leq a + y(1-a) \leq 1$, we have $0 \leq S(y) \leq -\log y$ as required fo r relative entropy.

- **Convexity in y:** Taking derivatives with respect to y

$$S'(y) = -\frac{1-a}{a+y(1-a)} \leq 0, \qquad S''(y) = \frac{(1-a)^2}{(a+y(1-a))^2} \geq 0. \tag{3.3}$$

  Thus $S(y)$ is decreasing and convex in the mixing parameter y (strictly convex if $s > 0$), consistent with the general theorem that $S(\rho\|\sigma)$ is convex in σ.

- **Monotonicity and concavity in s:** Treating S as a function of s:

$$\frac{\partial S}{\partial s} = \frac{(1-y)e^{-s}}{y+(1-y)e^{-s}} \geq 0, \qquad \frac{\partial^2 S}{\partial s^2} = -\frac{y(1-y)e^{-s}}{(y+(1-y)e^{-s})^2} \leq 0. \tag{3.4}$$

  So S increases with s but is concave in s, consistent with saturation at $-\log y$. A direct derivative gives the derivative of relative entropy in region size

$$K'(x) = \frac{\pi}{\sin^2(\pi x)}\left(\pi x - \frac{1}{2}\sin(2\pi x)\right) > 0 \qquad \text{for } x \in (0,1), \tag{3.5}$$

  since $\sin u < u$ for $u > 0$. Thus $K(x)$ (and therefore S) is increasing in x for $\alpha \neq \beta$, as expected from data processing.

Finally, we note that in the small interval limit $x \ll 1$ we have

$$K(x) = 1 - \pi x \cot(\pi x) = \frac{\pi^2}{3}x^2 + \frac{\pi^4}{45}x^4 + O(x^6). \tag{3.6}$$

Hence

$$S(y) = (1-y)\,(\alpha-\beta)^2\frac{\pi^2}{3}x^2 + O(x^4). \tag{3.7}$$

The relative entropy vanishes as x^2 for small intervals, as expected.

Alternative analytic continuation

An alternative analytic continuation follows from observing that since we are interested in small $n-1$, and the sum over $p = 1, \cdots, n-1$, we linearize our functions in p and discard higher order terms:

$$S_m(p) = -\frac{p}{2} + \frac{1-\cos(p\theta_m)}{4\sin^2(\theta_m/2)} = -\frac{p}{2} + O(p^2). \tag{3.8}$$

Expanding the g function in $n = 1+\epsilon$ results in

$$\log g(p;x) = p\,L(n,x) + O(p^2) = p\,L(n,x) + O(\varepsilon^2), \tag{3.9}$$

uniformly over the relevant range $p \in \{0, \ldots, n-1\}$. This reproduces the linear-in-p structure that one would obtain by keeping only the "smooth" term $S_m(p) \approx -p/2$ and effectively discarding the oscillatory $\cos(p\theta_m)$ piece at this order. While this results in a different analytic continuation than what we computed above, the $n \to 1$ limit gives the same result, and the same final expression (2.83) for relative entropy.

Pole-matching analytic continuation of $g(p;x)$ and $h(n-p,p;x)$

Here, we reformulate the analytic continuation of the building blocks $g(p;x)$ and $h(n-p,p;x)$ using the method used in the appendix of `arXiv:1508.03506`. The idea is to this: To find the analytic cotinuation of a meromorphic function $f_1(z)$, you find another entire function with the same poles and zeros $f_2(z)$ such that $f_1(z)/f_z(2)$ is bounded. Then, by Liouville's theorem it must be a constant C. This means that $Cf_2(z)$ is the analytic continuation of $f_1(z)$.

The basic building block and the definitions

Introduce

$$f_{n,m}(x) := \frac{\sin^2(\pi m/n)}{\sin\big(\pi(m+x)/n\big)\,\sin\big(\pi(m-x)/n\big)}\,, \qquad m \in \mathbb{Z}. \tag{3.10}$$

For integer $n \geq 2$ and $p \in \{1, \ldots, n-1\}$, the functions appearing in the correlator are

$$g(p;x) = \prod_{m=1}^{p-1} f_{n,m}(x)^{\,p-m}, \tag{3.11}$$

$$h(q,p;x) = \prod_{l=1}^{p}\prod_{k=1}^{q} f_{n,k+l-1}(x) = \prod_{r=1}^{n-1} f_{n,r}(x)^{\,w_r(p,q)}, \qquad q = n-p, \tag{3.12}$$

where $w_r(p,q)$ counts the number of pairs (k,l) such that $k+l-1=r$.

Poles of $g(p;x)$ as a function of x

From (3.11), all x-dependence sits in the denominator sines. For fixed m, the factor $\sin(\pi(m \pm x)/n)$ vanishes iff $m \pm x = \ell n$ for some $\ell \in \mathbb{Z}$, i.e.

$$x = \pm m + \ell n. \tag{3.13}$$

Hence the pole set of $g(p;x)$ is

$$\boxed{\mathrm{Pole}(g) = \Big\{\, x = \pm m + \ell n \;\Big|\; m = 1, \ldots, p-1,\ \ell \in \mathbb{Z} \Big\}.} \tag{3.14}$$

Moreover, each vanishing sine gives a simple pole, and the factor $f_{n,m}(x)$ is raised to power $(p-m)$. Therefore the *pole order* of $g(p;x)$ at $x = \pm m + \ell n$ equals $(p-m)$:

$$\boxed{\mathrm{ord}_{x=\pm m+\ell n}\, g(p;x) = p - m.} \tag{3.15}$$

Finally, $g(p;x)$ has no zeros as a function of x, since the numerator $\sin^2(\pi m/n)$ is x-independent.

A multiple-sine candidate and its divisor

We now introduce a candidate $g_{\mathrm{MS}}(p;x)$ built from multiple sine functions. Let $S_2(z|1,n)$ and $S_3(z|1,1,n)$ denote the double and triple sine functions (in a standard normalization), characterized by the shift relations

$$\frac{S_2(z+1\,|\,1,n)}{S_2(z\,|\,1,n)} = 2\sin\!\Big(\frac{\pi z}{n}\Big), \qquad \frac{S_3(z+1\,|\,1,1,n)}{S_3(z\,|\,1,1,n)} = \frac{1}{S_2(z\,|\,1,n)}. \tag{3.16}$$

Define the triangular-product package

$$T_p(z) := \frac{S_3(z+2\,|\,1,1,n)^{\,p}}{S_3(z+p+1\,|\,1,1,n)\,S_3(z+1\,|\,1,1,n)^{\,p-1}}, \tag{3.17}$$

and the candidate analytic continuation

$$g_{\mathrm{MS}}(p;x) := \frac{T_p(0)^2}{T_p(x)\,T_p(-x)}. \tag{3.18}$$

Claim (telescoping to the finite sine product). By iterating the shift relations (3.16), one checks that $T_p(z)$ is proportional to the triangular sine product

$$T_p(z) \;\propto\; \prod_{m=1}^{p-1} \sin\!\Big(\frac{\pi(m+z)}{n}\Big)^{p-m}, \tag{3.19}$$

where the proportionality factor is independent of z (it may depend on n,p). Consequently, (3.18) implies that for integer n, $g_{\mathrm{MS}}(p;x)$ reproduces the same pole lattice and the same pole orders as $g(p;x)$.

Zeros/poles of $g_{\mathrm{MS}}(p;x)$. From (3.19), $T_p(z)$ has zeros whenever $z = \ell n - m$ for $m = 1,\dots,p-1$, with zero order $(p-m)$. Therefore $g_{\mathrm{MS}}(p;x)$ has poles whenever $T_p(x)=0$ or $T_p(-x)=0$, i.e.

$$x = \ell n - m \quad \text{or} \quad x = -(\ell n - m) = m - \ell n \quad \Longleftrightarrow \quad x = \pm m + \ell n, \tag{3.20}$$

and the pole order at each such point equals $(p-m)$. In particular,

$$\boxed{\mathrm{Pole}(g_{\mathrm{MS}}) = \mathrm{Pole}(g), \qquad \mathrm{ord}_{x=\pm m+\ell n}\, g_{\mathrm{MS}}(p;x) = p-m.} \tag{3.21}$$

Like g, the function g_{MS} has no zeros as a function of x.

The ratio has no poles and is bounded

Define the ratio

$$R(x) := \frac{g_{\mathrm{MS}}(p;x)}{g(p;x)}. \tag{3.22}$$

Since g_{MS} and g have the same poles with the same orders and no zeros, all singularities cancel:

$$\boxed{R(x) \text{ is entire in } x.} \tag{3.23}$$

Moreover, both g and g_{MS} are even and n-periodic in x, hence so is R:

$$R(-x) = R(x), \qquad R(x+n) = R(x). \tag{3.24}$$

To argue boundedness, it suffices (by periodicity) to bound R on a single vertical strip $\{x = u + \mathrm{i}v : u \in [0, n]\}$. On such a strip, the original $g(p; x)$ obeys an exponential-type estimate as $|v| \to \infty$: for each fixed m,

$$\Big|\sin\Big(\frac{\pi(m \pm (u + \mathrm{i}v))}{n}\Big)\Big| \sim \frac{1}{2} e^{\pi|v|/n}, \qquad |v| \to \infty, \tag{3.25}$$

so

$$|g(p; u + \mathrm{i}v)| \sim C \exp\Big[-\frac{2\pi|v|}{n} \sum_{m=1}^{p-1} (p - m)\Big] = C \exp\Big[-\frac{\pi p(p-1)}{n}|v|\Big], \qquad |v| \to \infty, \tag{3.26}$$

with C independent of v (uniformly for u in compact sets away from poles).

On the other hand, the multiple sine functions S_2, S_3 have well-known large-imaginary asymptotics of the form

$$\log S_r(u + \mathrm{i}v \,|\, \boldsymbol{\omega}) = \pm \frac{\pi}{r!\,\omega_1 \cdots \omega_r} |v|^r \;+\; \text{lower powers in } |v| \;+\; O(1), \qquad |v| \to \infty, \tag{3.27}$$

so that in ratios such as (3.17) the leading polynomial growth cancels, and $T_p(z)$ has exponential type matching the finite sine product in (3.19). Consequently $g_{\mathrm{MS}}(p; u + \mathrm{i}v)$ has the same exponential type as $g(p; u + \mathrm{i}v)$, and hence their ratio $R(u + \mathrm{i}v)$ remains bounded as $|v| \to \infty$ uniformly in $u \in [0, n]$. Together with continuity on the compact region $\{u \in [0, n],\ |v| \le V\}$, we conclude:

$$\boxed{R(x) \text{ is bounded on } \mathbb{C}.} \tag{3.28}$$

Since R is entire and bounded, Liouville's theorem implies R is constant. Finally, $g(p; 0) = 1$ by definition and $g_{\mathrm{MS}}(p; 0) = 1$ by (3.18), so $R(0) = 1$ and thus

$$\boxed{g_{\mathrm{MS}}(p; x) = g(p; x) \quad \text{for all } x \text{ (at integer } n).} \tag{3.29}$$

This is precisely the "same poles + bounded ratio" strategy.

Corollary: $h(n-p, p; x)$ from a full-product identity

Using (3.12) with $q = n - p$ and the explicit multiplicities

$$w_r(p, n-p) = \begin{cases} r, & 1 \le r \le p, \\ p, & p < r \le n-p, \\ n-r, & n-p < r \le n-1, \end{cases} \tag{3.30}$$

one may rewrite $h(n-p, p; x)$ as

$$h(n-p, p; x) = \Big(\prod_{r=1}^{n-1} f_{n,r}(x)\Big)^p g(p; x)^{-2}. \tag{3.31}$$

The appendix identity of `arXiv:1508.03506` proves

$$\prod_{r=1}^{n-1} f_{n,r}(x) = \left(\frac{n\sin(\pi x/n)}{\sin(\pi x)}\right)^2, \tag{3.32}$$

again by the same poles/bounded ratio argument. Substituting into (3.31) yields

$$\boxed{h(n-p,p;x) = \left(\frac{n\sin(\pi x/n)}{\sin(\pi x)}\right)^{2p} g(p;x)^{-2},} \tag{3.33}$$

so the analytic continuation of h follows once that of g is fixed by the pole-matching method.

4 Information-theoretic interpretation